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# Neurotrophic isoindolinones from the fruiting bodies of *Hericium erinaceus*

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ABSTRACT

Four compounds, hericerin (**1**), isohericerinol A (**2**), *N*-de-phenylethyl isohericerin (**3**) and corallocin A (**4**) were isolated from the fruiting bodies of *Hericium erinaceus,* a lion's mane mushroom (Hericiaceae). Among them, isohericerinol A (**2**) was newly reported in nature. Further investigation of the neurotrophic effect of isolated compounds demonstrated that isohericerinol A (**2**) strongly increased the nerve growth factor (NGF) production in C6 glioma cells followed by corallocin A (**4**) and hericerin (**1**). Increased NGF production by these compounds promoted the neurite outgrowth in N2a neuronal cells. Western blot analysis also showed the increased protein expression of NGF, brain-derived neurotrophic factor (BDNF) and synaptophysin (SYP) in C6-N2a cells. Taken together, our present study characterized the neurotrophic constituents of *H. erinaceus*, which may support the potential use of memory improvement.

*Hericium erinaceum* (Hericiaceae) is an edible mushroom, which is mainly distributed in East Asia including Korea, Japan and China. It is commonly called as lion's mane mushroom and monkey's head mushroom due to the unique shape of white hair around the mushroom. It originally grows on dead trunks of hard woods and also cultivated these days. Traditionally, it has been used for the treatment of memory impairment and gastrointestinal diseases.[1–2] Previous phytochemical studies showed that the main constituents are erinacines, hericenones and polysaccharides, and diverse medicinal effects such as anti-inflammatory, anti-cancer, anti-bacterial and neurotrophic activities have been reported.[3–7]

Neurotrophins are important factors related to the growth, differentiation and death of nerve cells and nerve tissue in the nervous system.[8–9] The deficiencies of neurotrophins in neurons can cause neurodegenerative diseases such as Alzheimer's disease and Parkinson's disease. Hericenones and erinacines from this mushroom in particular have been shown to stimulate the synthesis of nerve growth factor (NGF) and activate the brain.[10–11]

For further finding an ingredient that could help the brain, isolation and structural characterization were conducted with chromatographic techniques and spectroscopic analysis in the present study. In addition, their stimulatory activity on neuronal cells was also evaluated.

Chromatographic separation of *n*-hexane fraction of *H. erinaceus*[12] resulted in the isolation of four compounds (Fig. 1) including one new compound.[13] Three known compounds were identified as hericerin (**1**), *N*-de-phenylethylisohericerin (**3**) and corallocin A (**4**) based on the spectroscopic analysis and by the comparison with the previous values.[14–16]

Compound **2** was obtained as yellow syrup and gave a pseudomolecular ion $[M + Na]^+$ by HRESIMS at 350.1361 (calculated for $C_{19}H_{21}NNaO_4$ 350.1363), consistent with a molecular formula of $C_{19}H_{21}NO_4$. The presence of a hydroxyl and γ-lactam groups was suggested by absorption bands at 3360 $cm^{-1}$ and 1644 $cm^{-1}$, respectively, in the IR spectrum. In the $^1H$ NMR spectrum, signals for the monosubstituted phenyl moiety at $\delta_H$ 7.19–7.27 (5H, m, H-4″-8″), an aromatic proton at $\delta_H$ 6.88 (1H, s, H-7), five methylenes at $\delta_H$ 2.98, 2.99, 3.66, 3.82 and 4.18 (each 2H, m, H-1′, 2″, 2′, 1″, 3, respectively) and a methoxy group at $\delta_H$ 3.85 (3H, s) were observed. The $^{13}C$ and HSQC

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spectra showed the corresponding peaks to two aromatic rings as twelve signals at $\delta_C$ 97.5–160.7, five methylenes including two aliphatic methylenes at $\delta_C$ 28.0 and 35.5, one oxymethylene at $\delta_C$ 62.5 and two methylenes adjacent to nitrogen at $\delta_C$ 49.3 (C-3) and 45.5 (C-1″), one carbonyl at $\delta_C$ 171.1 (C-1), and one methoxy group at $\delta_C$ 56.4. The HMBC correlations of H-3/C-4, C-7a; H-7/C-1, C-3a and comparison with NMR spectra of *N*-de-phenylethyl isohericerin (**3**) and corallocin A (**4**) suggested compound **2** as an isoindolinone derivative. Further HMBC correlations of H-1″/C-2″ and H-2″/C-3″, C-4″ and correlations of H-1′/C-2′ suggested the presence of phenylethyl and oxymethylene moieties in **2**. The positions of phenylethyl and oxymethylene moieties were determined to be N-2 and C-5, respectively, from HMBC correlations of H-1′/C-4, C-2′, H-2′/C-5 and H-1″/C-1, C-3. The HMBC correlation of $OCH_3$/C-6 suggested that the methoxy moiety was connected to C-6. On the basis of the obtained data, compound **2** was determined as 4-hydroxy-5-(2-hydroxyethyl)-6-methoxy-2-phenethylisoindolin-1-one (Fig. 1) and named isohericerinol A.[17]

Neurotrophins, which promote neuronal survival and neurite outgrowth, have a vital role in neurodegenerative diseases. In the central nervous system (CNS), astrocytes produce neurotrophins such as NGF and BDNF for protecting the neurons from neurodegenerative diseases.[18] Previous studies already showed the production of neurotrophin family in the cultured astroglial cell lines.[19–20] In addition, it has been reported that the natural products that increase the synthesis of neurotrophins could protect the neurons from injurious toxins, which released during different disease conditions. Thus, neurotrophic activity of the isolated compounds was evaluated by measuring the secretion of neurotrophic factors in C6 glioma and consequent effect on neurite outgrowth and N2a cells.[21] To determine the abundance of the released NGF in the culture medium, C6 glioma cells were treated with compounds **1**–**4** at concentrations of 10, 25 and 50 μM. Hericerin (**1**), isohericerinol A (**2**), and corallocin A (**4**) significantly increased NGF levels in culture medium compared to control. Especially, isohericerinol A (**2**), a new compound, showed the potent increase of the NGF level in a dose-dependent manner, whereas *N*-de-phenylethyl isohericein (**3**) showed little effect on NGF production (Fig. 2).

NGF production has been reported to be directly correlated with the differentiation and neurite outgrowth in the neuronal cells.[22] Thus, the

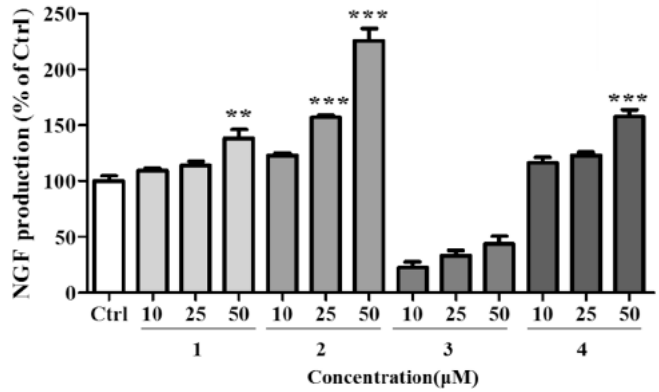


**Fig. 2.** Effect of compounds **1**–**4** from *H. erinaceus* on NGF level in C6 glioma cells. C6 glioma cells were treated with or without each compound at the concentrations 10, 25 and 50 μM for 24 h. After 24 h of incubation, conditioned medium was collected. The released NGF production was analyzed using enzyme-linked immunosorbent assay (ELISA) kit. All data are presented as the mean ± SEM (n = 3). $^{**}p < 0.01$ and $^{***}p < 0.001$ indicate significant differences compared to the untreated control group (Ctrl).

effect of increased NGF production on the neurite outgrowth was measured by assessing the length of neurites in N2a cells. After the treatment of compounds **1**, **2** and **4** at 25 μM, the gap of neurite length between 0 h and 24 h were measured. As shown in Fig. 3, neurite outgrowth was significantly increased by the treatment of hericerin (**1**) and isohericerinol A (**2**) up to 116.82 ± 1.97% and 122.82 ± 2.02%, respectively, which were higher than those of retinoic acid (112.08 ± 0.77%), a positive control. These results indicate that hericerin (**1**) and isohericerinol A (**2**) have potentiating effects on neurite outgrowth mediated by NGF synthesis.

Neurotrophins, especially NGF and BDNF, have been known to be secreted from astrocytes, which perform significant biological functions along with neurons in the brain: NGF plays a key role in cell differentiation, survival, development of neurite outgrowth and axonal guidance pathways, and protects the neurons from external stimuli and toxicity. BDNF is highly expressed in the CNS and is critically important

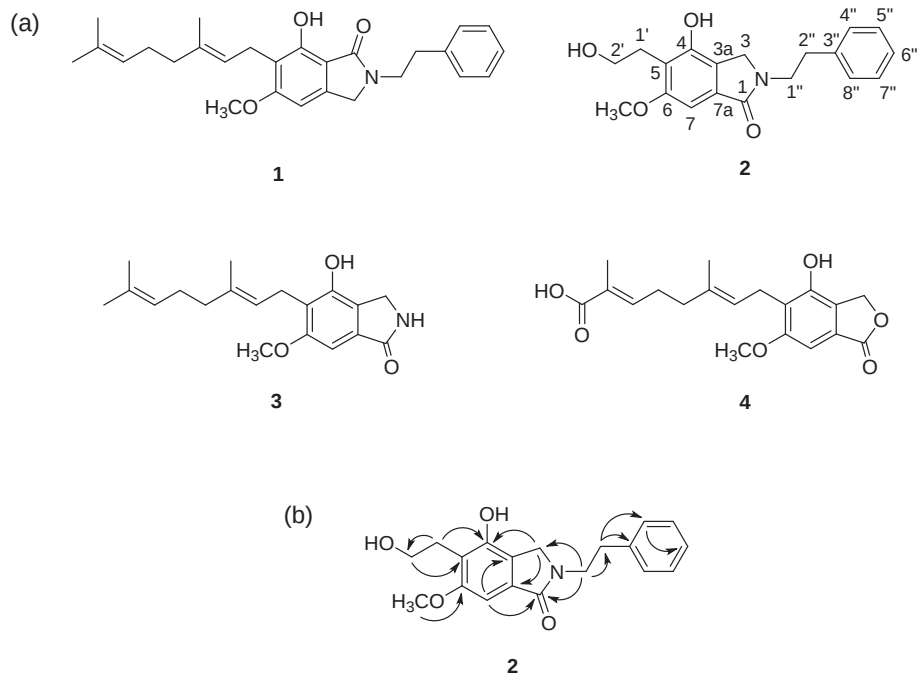


**Fig. 1.** (a) Chemical structures of compounds **1**–**4** from *H. erinaceus*; (b) Key HMBC (→) correlations of compound **2**.

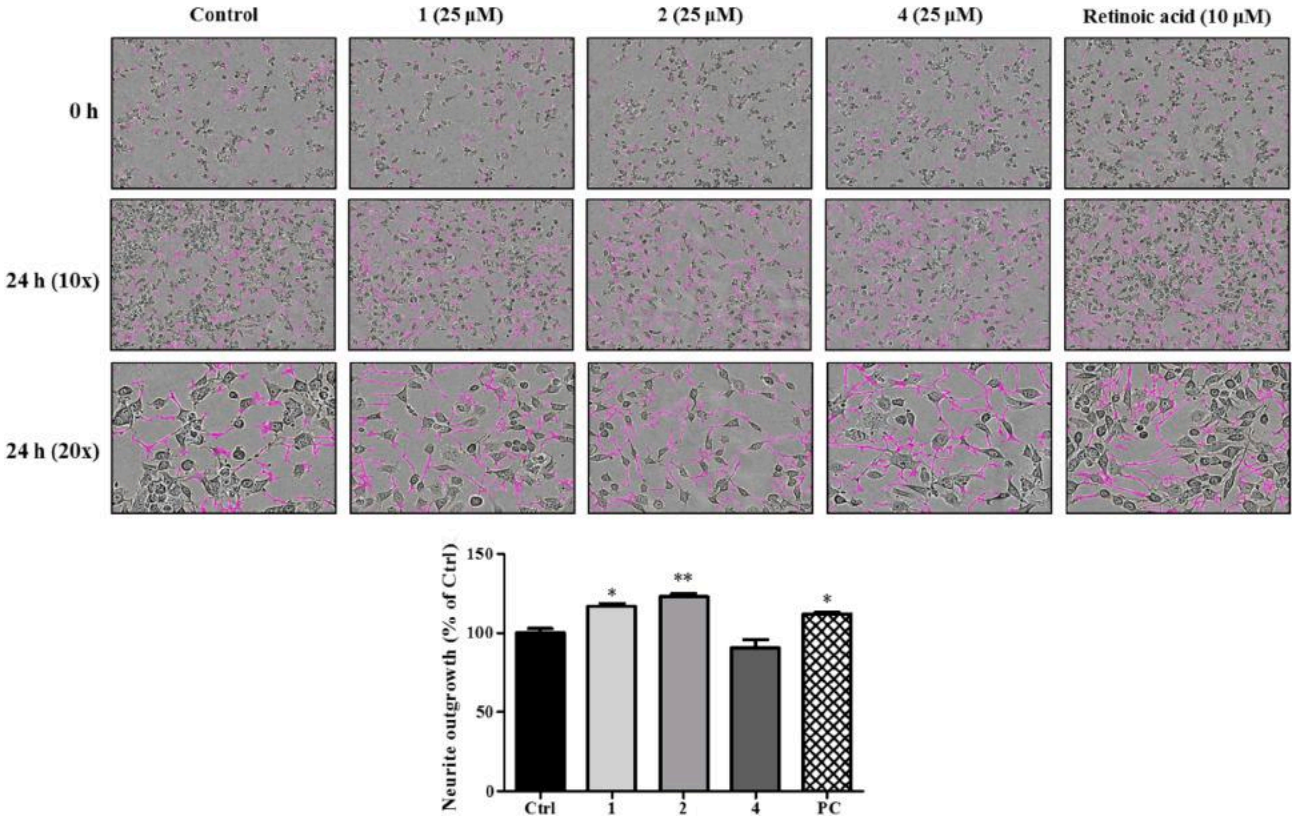


**Fig. 3.** The effect of compounds **1**, **2** and **4** on NGF-induced neurite outgrowth in N2a cells. N2a cells were treated with samples and incubated for 24 h. Sample treated N2a cells were incubated in the IncuCyte imaging system to analyze the neurite length and outgrowth. Scale bar = 200 μm. Retinoic acid (10 μM) was used as a positive control (PC). All data are presented as the mean ± SEM (n = 3). $^{*}p < 0.05$ and $^{**}p < 0.01$ indicate significant differences compared to the untreated control group (Ctrl).

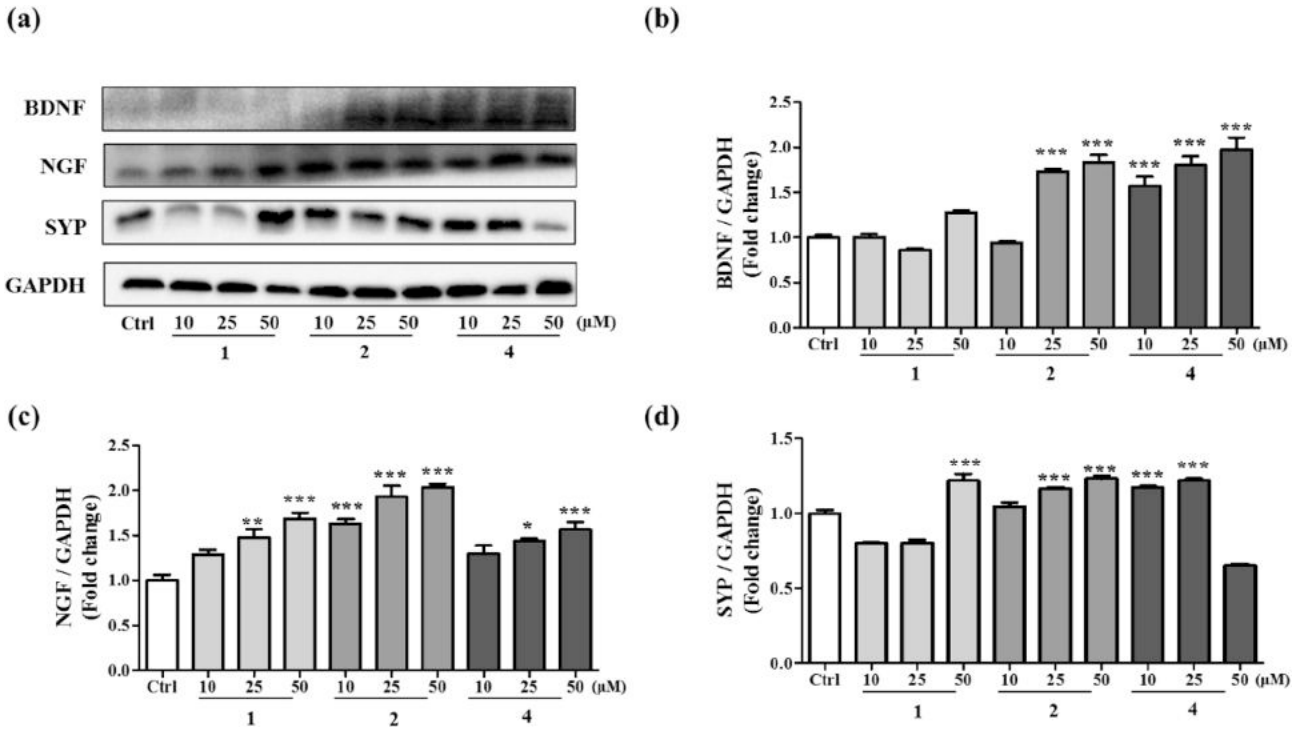


**Fig. 4.** The effect of compounds **1**, **2** and **4** on expression of NGF, BDNF and SYN protein in C6-N2a cells. N2a cells treated with C6 conditioned medium, after 24 h of incubation cells were collected. (a) Protein expression of BDNF, NGF, or SYP. (b-d) Relative band intensity of BDNF (b). NGF (c), or SYN (d). GAPDH was used as a loading control. All data are presented as the mean ± SEM (n = 3). $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$ indicate significant differences compared to the untreated control group (Ctrl).

for survival of neurons.[23] To explore the astrocyte and neuron network of isolated compounds, C6 glioma cells were treated with isolated compounds at the concentrations of 10, 25 and 50 μM to obtain conditioned medium, which was eventually transferred to N2a cells (C6-N2a). As shown in Fig. 4, the expressions of NGF protein were significantly upregulated by hericerin (**1**), isohericerinol A (**2**) and corallocin A (**4**) at the concentrations of 10, 25 and 50 μM. In addition, isohericerinol A (**2**) and corallocin A (**4**) also increased the expression of BDNF protein. Synaptophysin (SYP) which occurred in presynaptic vesicles of neurons, is widely used as a marker for synaptic integrity.[23] There was significant changes in the expressed level of SYP after treatment of each compound. These results suggested that hericerin (**1**), isohericerinol A (**2**) and corallocin A (**4**) from *H. erinaceus* might promote synaptic plasticity.

In summary, we isolated one new isoindolinone derivative named isohericerinol A (**2**) together with three known compounds, hericerin (**1**), *N*-de-phenylethyl isohericerin (**3**) and corallocin A (**4**) from the fruiting bodies of *H. erinaceus*. Compounds **1** and **2** significantly increased the neurite outgrowth by NGF synthesis. In addition, the levels of neutrophins including NGF and BDNF together with SYP were also upregulated by these compounds. Conclusively, the present study suggested that compounds **1** and **2** could promote neuronal differentiation and neurite outgrowth, which may be beneficial for preventive and therapeutic uses in neurodegenerative diseases.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bmcl.2020.127714.

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12 H. erinaceus were cultivated in the C&G Agricultural Association (Sejong, Korea). Briefly, strain of H. erinaceus obtained from Jeollanamdo Agricultural Research Institute (Naju, Korea) was maintained on the potato dextrose agar (PDA 200 g/l diced potatoes; 20 g/l glucose; 15 g/l agar) medium for regular subculture. They were growing media composed of sawdust (SD) as a base substrate with 49% together with cottonseed hull (CH, 19%), rice bran (RB, 7%), cottonseed meal (CM, 12%) and beet (BT, 13%) with the control of humidity, temperature, ventilation and light (See Supplementary data).
13 The dried fruiting bodies of H. erinaceus (3.8 kg) were pulverized and extracted twice with EtOAc. The EtOAc extract (180 g) was suspended in a mixture of CH2Cl2-MeOH-H2O (10:9:1) and partitioned with n-hexane to give n-hexane fraction. Compounds 1 - 4 were isolated from the n-hexane-soluble fractions using column chromatography (See Supplementary data). A voucher specimen (CBNU2017-HEF) was deposited in the herbarium of College of Pharmacy, Chungbuk National University.
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17 Isohericerinol A (2) Light yellow syrup; \left[ \alpha \right]_D^{25}+16 (c 0.01 CH3OH); IRmax 3340, 1644 cm-1; UV (MeOH) λmax 212, 259 nm; 1H NMR (CD3OD, 500 MHz): δ 7.22-7.27 (4H, m, H-4″, 5″, 7″, 8″), 7.19 (1H, m, H-6″), 6.88 (1H, s, H-7), 4.18 (2H, s, H-3), 3.85 (3H, s, OCH3), 3.82 (2H, overlap, H-1″), 3.66 (2H, t, J = 7.0 Hz, H-2′), 2.99 (2H, overlap, H-2″), 2.98 (2H, overlap, H-1′) ppm; 13C NMR (CD3OD, 125 MHz), δ 171.1 (C-1), 160.7 (C-6), 152.4 (C-4), 140.2 (C-3″), 132.8 (C-7a), 129.9 (C-4″, 8″), 129.7 (C-5″, 7″), 127.6 (C-6″), 122.6 (C-3a), 119.7 (C-5), 97.5 (C-7), 62.5 (C-2′), 56.4 (OCH3), 49.3 (C-3), 45.5 (C-1″), 35.5 (C-2″), 28.0 (C-1′); ESI-MS (positive node) m/z 350 [M+Na]+; HRESIMS (positive node) m/z 350.1361 (calculated for C19H21NNaO4 350.1363).
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